# 1 Introduction

## Large-Scale Structure of the Universe

**Definition:** The large-scale structure of the universe describes the hierarchical organization of matter, from planetary systems to structures extending over billions of light-years.

**1. Solar System**

- Sun's radius: $6.96 \times 10^5$ km $\simeq 109\, R_\oplus$ (earth's radius)
- Diameter (to Neptune): $\sim 9 \times 10^9$ km
- Bound by the Sun's gravity

**2. Star Clusters** (gravitationally bound)

- *Open Clusters:*
  - $10^2$–$10^3$ stars
  - Size $\lesssim 30$ light-years
  - Young and loosely bound
- *Globular Clusters:*
  - $10^5$–$10^6$ stars
  - Size $\sim 100$–$200$ light-years
  - Old and dense (e.g. Omega Centauri)

**3. Galaxies**

- Composed of stars, gas, dust, stellar remnants, and dark matter
- Stellar content: $10^7$–$10^{12}$ stars
- Size: $3 \times 10^3$–$3 \times 10^5$ light-years
- Observable universe contains hundreds of billions of galaxies

**4. The Milky Way**

- Spiral galaxy
- $\sim (1$–$4) \times 10^{11}$ stars
- Diameter $\sim 9 \times 10^4$–$10^5$ light-years

**5. Galaxy Associations**

- *Galaxy Groups:*
  - Up to $\sim 50$ galaxies
  - Example: Local Group ($\sim 50$–$80$ galaxies)
- *Galaxy Clusters:*
  - $10^2$–$10^3$ galaxies
  - Size: 1–15 million light-years

- Example: Virgo Cluster

- *Galaxy Superclusters:*
  - Collections of clusters and groups
  - Not gravitationally bound as a whole
  - Expand with cosmic expansion

**6. Filaments and Walls**

- Largest known structures in the universe
- Networks of galaxies and dark matter surrounding cosmic voids
- Length scales: $10^7$–$10^{10}$ light-years
- Examples:
  - Super-Virgo Wall ($\sim 10^8$ light-years)
  - Hercules–Corona Borealis Great Wall ($\sim 10^{10}$ light-years)

**Hierarchy Summary:**

Solar Systems → Star Clusters → Galaxies → Groups → Clusters → Superclusters → Filaments and Voids

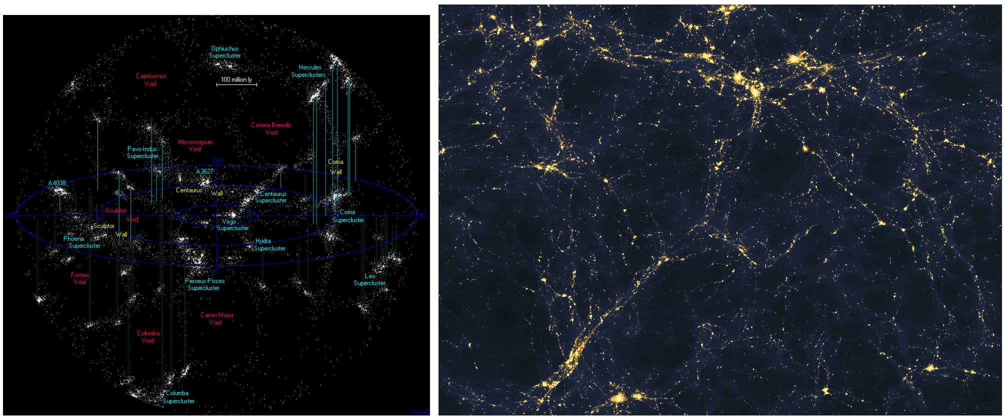


Figure 1: Galaxy superclusters and voids (left). Galaxy filaments (right).

## 1.1 Units

### Units in Cosmology and Particle Physics

Due to the enormous range of scales involved, conventional units such as km and kg are impractical. Instead, specialized astrophysical and natural units are used.

**Cosmology Units**

**Solar Units**

- Solar mass:
$$M_\odot = 1.99 \times 10^{33} \text{ g}$$
- Solar luminosity:
$$L_\odot = 3.85 \times 10^{33} \text{ erg s}^{-1}$$

**Distance Units**

- Astronomical Unit (AU):
$$1 \text{ AU} = 1.496 \times 10^{13} \text{ cm}$$
- Light-year (ly):
$$1 \text{ ly} = 9.46 \times 10^{17} \text{ cm}$$
- Parsec (pc): distance at which 1 AU subtends an angle of 1 arcsecond,
$$1 \text{ pc} = 3.26 \text{ ly}$$
- Megaparsec (Mpc):
$$1 \text{ Mpc} = 10^6 \text{ pc}$$

**Remark:** Cosmology is typically concerned with length scales $\gtrsim 100$ Mpc.

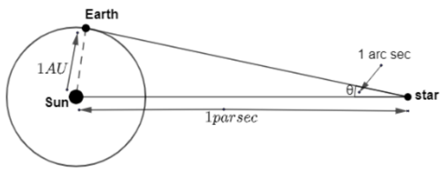


Figure 2: Definition of one parsec.

**Natural Units**

Fundamental constants frequently appearing in particle physics are

- Speed of light:
$$c = 3 \times 10^8 \text{ m s}^{-1}$$
- Reduced Planck constant:
$$\hbar = 1.055 \times 10^{-34} \text{ J s} = 6.58 \times 10^{-16} \text{ eV s}$$
- Boltzmann constant:
$$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1} = 8.62 \times 10^{-5} \text{ eV K}^{-1}$$

(Using $1 \text{ eV} = 1.602 \times 10^{-19}$ J.)

**Natural unit system:**
$$c = \hbar = k_B = 1$$

In this system, all quantities are expressed in powers of energy (eV).

- Time:
$$1\ \text{s} = 1.52 \times 10^{15}\ \text{eV}^{-1}$$
- Length:
$$1\ \text{cm} = 5.06 \times 10^{4}\ \text{eV}^{-1}$$
- Temperature:
$$1\ \text{K} = 8.62 \times 10^{-5}\ \text{eV}$$

**Energy scales:**
$$1\ \text{GeV} = 10^{9}\ \text{eV}$$

**Planck Units:**

Planck units are based on four fundamental constants: $c$, $G$, $\hbar$, and $k_B$. The key Planck units are:
$$t_{\text{Pl}} = 5.39 \times 10^{-44}\,\text{s}, \quad \ell_{\text{Pl}} = 1.16 \times 10^{-33}\,\text{cm}, \quad m_{\text{Pl}} = 2.17 \times 10^{-5}\,\text{g}, \quad T_{\text{Pl}} = 1.14 \times 10^{32}\,\text{K}.$$
In natural units, Planck mass is:
$$m_{\text{Pl}} = 1.22 \times 10^{28}\,\text{eV}.$$

## 1.2 A Brief History

The universe is expanding, with galaxies moving away from each other. This implies that the universe was once hotter and denser, in a state known as the *Big Bang*. The theory explaining this is the *Big Bang Theory*. However, there are common misconceptions:

- "The universe started from the Big Bang" – The Big Bang refers to a hot, dense state, not a moment of creation.
- "The Big Bang was an explosion at some point in space" – The Big Bang was a universal event, occurring everywhere in space, not at a specific point.
- "The universe is expanding into open space" – The universe is not expanding into anything; space itself is expanding.

As events during the Big Bang are beyond current physics, time is often referenced as $t = 0$. A more practical way to track the universe's history is through temperature. Key events include:

- **Planck Time:** $t \sim 10^{-43}\,\text{s}$, $T = 10^{19}\,\text{GeV}$: Physics breaks down; quantum gravity theories become relevant.
- **Inflationary Stage:** $t \sim 10^{-37}\,\text{s}$ to $10^{-32}\,\text{s}$, $T = 10^{17}\,\text{GeV}$: Exponential expansion, somewhat speculative with no experimental evidence.
- **Quark-Gluon Plasma:** $t \sim 10^{-35} - 10^{-5}\,\text{s}$, $T \sim 1 - 100\,\text{GeV}$: Dominated by subatomic particles.
- **QCD Phase Transition:** $t \sim 10^{-5}\,\text{s}$, $T \sim 100 - 300\,\text{MeV}$: Quarks bind to form hadrons.
- **Big Bang Nucleosynthesis:** few seconds to $t \sim 3\,\text{minutes}$, $T < 100\,\text{MeV}$: Formation of light elements; provides evidence for the Big Bang. Very well understood.
- **Matter-Radiation Equality:** $t \sim 10^{4-5}\,\text{years}$, $T \sim 3\,\text{eV}$: The universe becomes matter-dominated.
- **Recombination:** $t \sim 400,000\,\text{years}$, $T \sim 1\,\text{eV}$: First hydrogen atoms form, and the universe becomes transparent, creating the CMB.
- **Formation of Stars and Galaxies:** After 400 million years, stars and galaxies form. The solar system appeared 9 billion years after the Big Bang. Today, the universe is 13.7 billion years old, with a temperature of $T = 2.7\,\text{K}$.

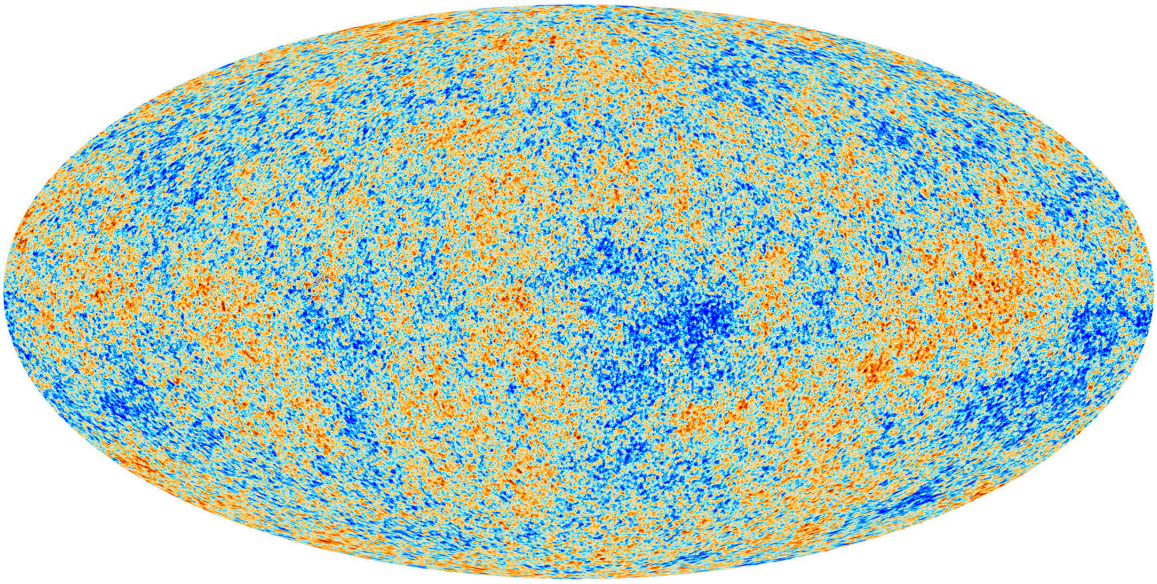

Figure 3: The cosmic microwave background (CMB) courtesy the ESA's Planck mission.

### 1.3 Olber's Paradox:

Until the early 20$^{th}$ century, it was believed that the universe was infinite, eternally old, and static. These assumptions led to a paradox. The night sky is dark, with stars and galaxies scattered across it. If the universe is infinite, the distribution of stars and galaxies would also extend infinitely. And if the universe is eternally old, so would the stars and galaxies be. Therefore, if we extend our line of sight infinitely, we should eventually encounter a star or galaxy, meaning the sky should not appear dark. In fact, assuming all stars are as bright as the Sun, calculations suggest that the sky should be as bright as the Sun's surface. However, this contradicts our observations. This is known as Olber's Paradox.

The resolution of Olber's Paradox requires rejecting one or more of the initial assumptions:

- **Finite Age of the Universe:** The universe is not eternally old. It has a finite age, approximately 13.7 billion years. This means that stars have only existed for a finite time, and the light from far-off stars has not had enough time to reach us. As a result, we can only observe stars whose light has had time to travel to Earth.
- **Expanding Universe:** The universe is not static; it is expanding. As the universe expands, galaxies move away from us, and their light becomes redshifted. This makes their light less visible, reducing the number of stars and galaxies whose light can reach us.
- **Intervening Absorption:** The universe contains non-luminous objects such as dust, gas, and other matter that may block our line of sight to distant stars, preventing us from seeing all of them.

Thus, the combination of the universe having a finite age, the expansion of the universe, and the absorption of light by cosmic materials resolves Olber's Paradox. Rather than an infinitely bright sky, we observe a dark night sky with scattered stars.

### 1.4 Isotropy and homogeneity

The study of the universe is grounded in the *cosmological principle*, which asserts that *on sufficiently large scales, the universe is homogeneous and isotropic.* Homogeneity implies that there is no privileged location in the universe; it appears the same from any point of view. Isotropy means that all directions from a given point are equivalent. While the cosmological principle is not directly proven, observational evidence, such as the uniformity of the cosmic microwave background (CMB) radiation, strongly supports it. Additionally, $N$-body simulations of the universe, based on the $\Lambda$CDM model, naturally result in an isotropic and homogeneous universe.

## 1.5 Hubble's Law

In 1929, Edwin Hubble observed that the light from distant galaxies is redshifted, and this redshift is proportional to the distance of the galaxy. The redshift, $z$, is defined as:

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{em}}}{\lambda_{\text{em}}},$$

where $\lambda_{\text{em}}$ is the wavelength of the emitted light, and $\lambda_{\text{obs}}$ is the wavelength observed on Earth. If $r$ represents the distance to a galaxy, Hubble's observation implies:

$$z \propto r.$$

If the redshift is attributed to the Doppler effect, the shift should be proportional to the relative velocity between the observer and the galaxy:

$$z \propto v.$$

Thus, we have the relationship:

$$v \propto r, \quad \text{or} \quad \boxed{v = H_0 r},$$

where $H_0$ is the constant of proportionality, known as the Hubble constant. This equation states that galaxies are receding from us with velocities proportional to their distance. This relationship is known as Hubble's Law. In Figure 5, the receding velocities are plotted against the distance to the galaxies.

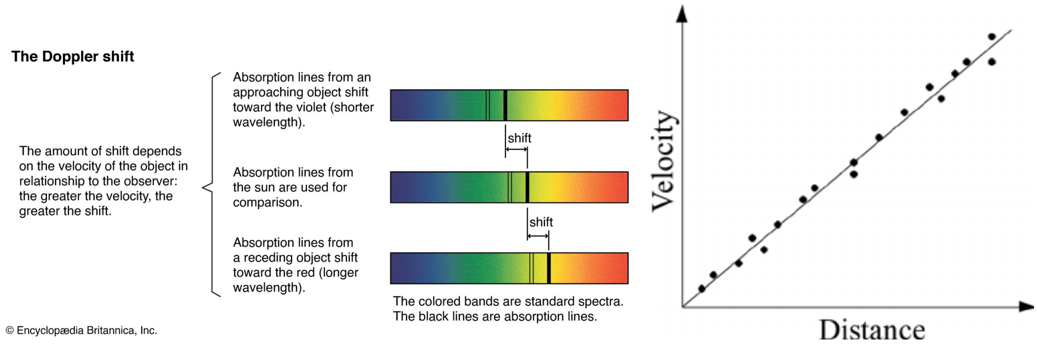


Figure 4: Hubble's observation: galaxies receding at velocities proportional to distance.

## 1.6 Universality of Hubble constant

Since the universe is expanding as well as it is homogeneous and isotropic, the Hubble constant is same for all observers in this universe. Imagine three galaxies located at positions $\vec{r}_1$, $\vec{r}_2$, and $\vec{r}_3$, forming a triangle in space. The distances between these galaxies are $r_{12} = |\vec{r}_1 - \vec{r}_2|$, $r_{23} = |\vec{r}_2 - \vec{r}_3|$, and $r_{31} = |\vec{r}_3 - \vec{r}_1|$. According to the cosmological principle, the galaxies recede from each other in a homogeneous manner, meaning that the shape of the triangle formed by these galaxies remains constant over time. In other words, the relative distances between the galaxies scale with a common factor, which we denote as $a(t)$. Therefore, the relative distances at time $t$ are related to those at time $t_0$ as:

$$r_{12}(t) = a(t) r_{12}(t_0)\,, \quad r_{23}(t) = a(t) r_{23}(t_0)\,, \quad r_{31}(t) = a(t) r_{31}(t_0)\,.$$

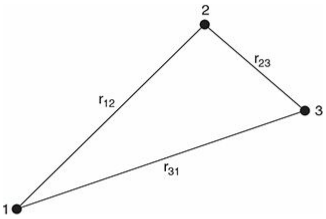


Figure 5: Hubble's observation: galaxies receding at velocities proportional to distance.

and receding speeds are

$$v_{12}(t) = \frac{\dot{a}}{a} r_{12}(t)\,, \quad r_{23}(t) = \frac{\dot{a}}{a} r_{23}(t)\,, \quad r_{31}(t) = \frac{\dot{a}}{a} r_{31}(t)\,. \tag{1}$$

So, any two points in the universe is receding away from each other with speed that is proportional to their distance of separation and the proportionality constant is

$$H_0 = \frac{\dot{a}}{a}$$

is universal. This was the first direct evidence of the expansion of the universe.

The Hubble constant $H_0$ has a dimension of $t^{-1}$. Hubble's own measurement of this constant was about seven times larger than present measurements using CMB data and using Supernova (explosion of a star) data. Though they yield slightly different values we will use

$$H_0 = (72 \pm 7)\ km\ s^{-1} Mpc^{-1}\,. \tag{2}$$

In literature a dimensionless Hubble constant $h$ is more often used

$$H_0 = 100h\ km\ s^{-1} Mpc^{-1} \tag{3}$$

where $h = 0.72 \pm 0.007$.

The Hubble constant is not a constant in time – throughout the different stages of the universe it had different values. The value quoted above is the one at the present epoch – and thats why I have put 0 in the suffix.

As $H_0$ has the dimension of inverse time, $H_0^{-1}$ is approximately the age of the universe. If the universe is expanding any two galaxies were closer in the past. Let's assume that there is no force in the universe to accelerate of decelerate the relative motion of galaxies and they recede at constant speed. So, two galaxies have taken a time

$$t_0 = \frac{r}{v} = \frac{r}{H_0 r} = H_0^{-1}\,,$$

to be separated by distance $r$. This is approximately the age of the universe. With $H_0 = 72 km/s/Mpc$ one gets $H_0^{-1} \approx 14$ billion years as the universe's age.